Settimana: 21

Materia: Matematica
Classe: 5A
Data: 02/03/26

Pag 1972 n 427:

$$\int x^2 \sin x \, dx \overset{\text{Per Parti}}{=}$$

$$\int f'g = fg - \int fg'$$

Goal: abbassare il grado di x^2

$$-x^2\cos x + \int 2x \cos x \, dx =$$

$g = x^2$ $\quad g' = 2x$

$f = -\cos x$ $\quad f' = \sin x$

$$-x^2\cos x + 2x\sin x - \int 2\sin x \, dx =$$

$$\boxed{-x^2\cos x + 2x\sin x + 2\cos x + c}$$

$g = 2x$ $\quad g' = 2$

$f = \sin x$ $\quad f' = \cos x$

$2\sin x\cos x = \sin 2x$

438: $$\int 8x \sin x \cos x \, dx = \int 4x \sin 2x \, dx$$

$2x = t$
$2dx = dt$

$$= \int t \sin t \, dt = -t\cos t + \int \cos t \, dt$$

$g = t$ $\quad g' = 1$

$f = -\cos t$ $\quad f' = \sin t$

$$= -t\cos t + \sin t + c$$

$$= \boxed{-2x\cos 2x + \sin 2x + c}$$

$$\int f'g = fg - \int fg'$$

Hard: $$\int \cos^2 x \, dx = \int \cos x \cdot \cos x \, dx =$$

$f = \sin x$ $\quad f' = \cos x$
$g = \cos x$ $\quad g' = -\sin x$

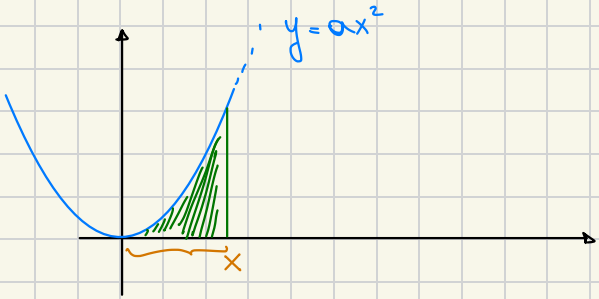
$$\sin x \cos x + \int \sin^2 x \, dx$$

$$\int \cos^2 x \, dx = \sin x \cos x + \int \underbrace{\sin^2 x}_{1-\cos^2 x} \, dx$$

$$\underbrace{\int \cos^2 x \, dx}_{R} = \sin x \cos x + \int 1 \, dx - \underbrace{\int \cos^2 x \, dx}_{R}$$

$$R = \sin x \cos x + x - R$$

$$\cancel{2}R = \boxed{\frac{\sin x \cos x + x}{2} + c = \int \cos^2 x \, dx}$$



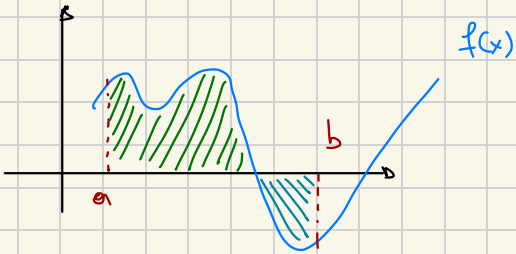
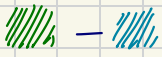
Area Verde $= \frac{ax^3}{3}$

$$\int ax^2 \, dx = \frac{ax^3}{3}$$

Teorema: Data $f : [a;b] \longrightarrow \mathbb{R}$ funzione integrabile, l'integrale

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

con F primitiva di f è l'area sottesa dal grafico e dell'asse x



$$\int_a^b f(x) \, dx =$$